

Architecture Selection & Justification

System: Webhook Ingestion & Retry Mechanism Hub

Architecture Selection: We chose a Microservices Architecture for the Webhook Ingestion & Retry Mechanism Hub.

Reason 1 – Independent processing:

The ingestion, delivery, retry, queue, monitoring, and dead-letter functions are separated into focused services. This makes each function easier to develop, maintain, and modify.

Reason 2 – Scalability and fault isolation:

Webhook traffic and delivery workload can increase independently. Queue-based processing supports scaling and helps isolate failures in delivery or retry handling.

Security Advantage:

The API Gateway provides a controlled entry point where webhook requests can be authenticated and validated before events enter the processing pipeline.

Performance Benefit:

The inbound queue decouples ingestion from delivery. Events can be accepted quickly and processed asynchronously by delivery workers, helping during bursts of incoming events.

Conclusion: The selected architecture separates webhook ingestion, asynchronous processing, delivery, retry handling, persistence, and monitoring into clear components with defined interfaces and dependencies.